

2026-08-14 Construction of the analysis sample

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Introduction

- I will introduce the construction of the analysis samples in this report.
 - The big decision is the construction of job postings sample.
- There are two linked but not directly linked samples for this project:
 - ① The sample of new hires: Its construction is intuitive and definitive.
 - ② The sample of job postings: It is not so straightforward and we basically stopped here last week.
- In what follows, I will talk about the important decisions in the process of constructing these two analysis samples, using the BioPharm industry in US as an example.

Outline

- ① Two analysis samples
- ② A weighted sample of job postings
- ③ Next steps

The final sample of focal new hires (straightforward and intuitive)

- The construction process:
 - ① Keep employment spells in the focal industry-occupation (e.g., scientists in the BioPharm industry).
 - ② Further retain employment spells starting from Jan 2021 to Dec 2023.
 - ③ Exclude employment spells with missing geography or job title information.
 - ④ Exclude internship positions.
 - ⑤ Retain one employment spell within each user-company cell ([the final sample of focal new hires](#)).
- Interpretations:
 - It contains one row per user-company cell; the same user can appear at multiple companies.
 - Internal transfers into a focal occupation are kept in the sample deliberately.
 - For example, if a lab assistant (which is not a focal occupation) gets promoted as a scientist, then I keep only his/her scientist employment spell and count as a new hire.
 - Whether a company hires internally or externally is a part of its hiring practices. This sample will be useful to determine to what extent a company is using network-based hiring practices.

The sample of LinkedIn job postings (tricky)

Background information about the raw job postings data:

- Each row is one unique job posting.
- Most rows have non-missing job titles and ONET occupation classifications.
- The variable `description` stores the full job advertisement texts scraped by Revelio.
 - This variable could be missing even if the job title variable is not missing.
 - This variable could be nonempty but meaningless. [▶ Examples](#)
 - I will call the job descriptions without enough information to extract any variable from them as **bad/meaningless/uninformative** interchangeably.
 - Repetition of the same description texts is also an issue.
 - There are 395,488 job postings in the US. However, there are only 148,222 distinct job descriptions.¹
 - Among the 1,910 US companies, only 335 of them have no repeated job description texts.²
 - This could suggest: The degree of selectivity measured from job postings may be better interpreted as a “fixed type”.

¹ $395,488/148,222 = 2.67$.

² $335/1,910 = 17.5\%$.

What do we want to measure from job postings?

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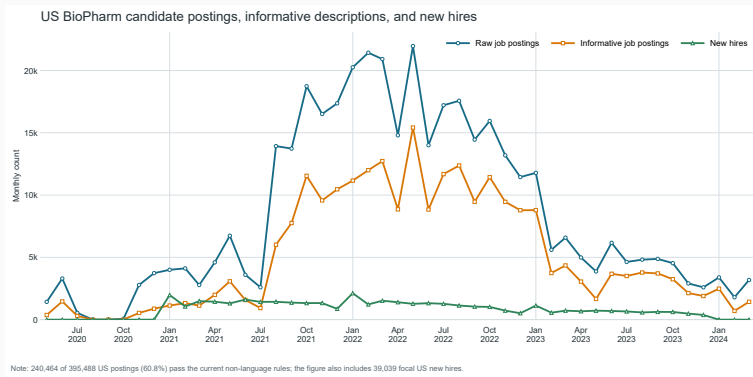
- **The degree of selectivity: Education/experience/skill/qualification requirements.**
 - I didn't find any economics papers which systematically measure selectivity from job advertisements.
 - Deming and Kahn (2018) is the most relevant paper. They focus on 10 general skills and measure them using a dictionary-based keyword matching method. This is still different from our purpose.
 - Some NLP and computational linguistics papers talk about the workflow to extract (specific) skills from job postings: e.g., Green et al. (2022), Krüger et al. (2025), Nguyen et al. (2024), and Zhang et al. (2022).
- Disclosed screening procedures: Tests/interviews/other screening procedures.
- Optional: Amenities (remote work, flexibility, culture, insurance, etc).

Why can't we simply group all job postings within a company-year?

It is tempting that we start from all companies in the BioPharm industry for a given year and keep all job postings (whether meaningful or meaningless) for scientists, and construct variables measuring selectivity using a simple average across all job postings. Two reasons why this may not be a good idea.

- The difference between two “times”.
 - There is a natural lag between a new hire joins the company (the start time) and the company posts a job advertisement (the post time), and this lag is unobservable.
 - In the end, we need to combine the selectivity measures from job postings and new hires' career trajectories. However, it makes little sense to match firms' 2021 selectivity measure to its 2021 cohort of new hires.
- Not all job postings should be weighted equally.
 - Ghost hires:
 - According to a Revelio's report ([link](#)): “The ratio of hires per job posting in the US has fallen from 0.75 in 2018 to below 0.5 in 2023, meaning fewer than one in two US job postings now lead to an actual hire.”
 - We should focus on the real job postings that potentially lead to new hires.
 - Other cases: evergreen job posts, repetitive job posts, etc.

There are substantially more candidate job postings than focal new hires.



- Green line: 39,039 focal new hires (BioPharm, US, new scientists from 2021 to 2023).
- Blue line: 395,488 candidate job postings (from the same companies hiring scientists). [▶ Construction](#)
- Orange line: 240,464 **meaningful** candidate job postings (meaningful = we can construct selectivity measures from the job description texts). [▶ Construction](#)
- Implications: We need to restrict the sample of job postings somehow in a reasonable way.

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A more reasonable way to construct the sample of job postings

- For each new hire, we match him with all potential job postings from the same company (e.g., the companies's job postings for scientists 1 to 9 months before his start time).
 - The matched job postings can have different weights (e.g., we can assign a higher weight to a job posting that is closer to the new hire's start date).
 - The sum of weights across all job posting matched to one new hire will always be 1.
 - For each new hire, using all mapped candidate job postings, we can construct a weighted average level of selectivity facing by each new hire.³
- Interpretations:
 - For each new hire, there is one synthetic job posting. This synthetic job posting is essentially the weighted average of multiple real-world job postings.
 - In this way, we try to mimic the hiring environment (i.e., the ex ante level of selectivity before he joins the company) a new hire may face during his job search process.
 - A ghost job posting that doesn't lead to any new hires will not be included.

³An implicit assumption here is that we can only use the meaningful job advertisements to construct this average level of selectivity. We can construct variables measuring the share of meaningless job postings, yet the measure of selectivity has to be calculated only within the meaningful job postings.

Potential robustness checks for the job postings sample

- I have derived a weighting scheme based on Poisson assumptions. [▶ A potential weighting scheme](#)
 - The key parameter there is the average number of months a job posting takes to fill in.
- We should do a bunch of robustness checks on the construction of job postings sample.
 - Using the subsample where we can confidently match a new hire to one single job posting.
 - Using different weights (maybe equal weights).
- Final comments: I would say, the construction of the analysis sample (especially for the job postings sample) is more like an art for our project.
 - The sample of new hires are linked to the sample of job postings, but there is no direct merge key that we can rely on.
 - In my opinion, a weighted sample of job postings based on the sample of new hires is better to keep all job postings and calculate a simple average of selectivity indicators across all job postings.
 - In this way, we focus only on those job postings that could potentially lead to realized new hires, i.e., remove the ghost job postings.

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Next steps

- A deterministic, rule-based algorithm to identify the meaningful job postings.
 - The output of this step is a sample of meaningful job postings, which will be fed into next step to extract selectivity measures.
 - I am currently working on this (with a first-round algorithm).
 - Some human validation is needed to make sure this algorithm is robust and reasonable so that we won't lose false negative (meaningful job postings identified to be meaningless) cases.
- LLM or other machine learning models to construct a series of indicators measuring the level of selectivity with the job postings.
 - Surprisingly, I don't think there is any economics paper trying to build a reasonable pipeline for this task.
 - This could be one of our contributions too – the way to measure selectivity from job advertisements.

More on the analysis sample: At which level of aggregation

- After determining the sample of new hires and job postings, we need to determine the level of aggregation. That is to say, is our revealed hiring practice measure at company level, or company-year level, or company-state level, or company-state-year level.
- This is partly a data-driven process, because within each cell, we need a minimum number of new hires to make our measurement meaningful.
- This will also be based on implications/validation tests we want to have. In some earlier discussion, we talk about using mergers/IPOs as shocks to examine if companies change their hiring practices after these events, then the level of aggregation has to be at least company-year level.

Outline

④ Appendix: Supplementary materials

⑤ Appendix: References

Bad job postings: No information at all

► Back

```
print(df[df["job_id"] == 2149963273]["description"].to_list())  
# ['N/A']
```

```
print(df[df["job_id"] == 2322300942]["description"].to_list())  
# ['\u200b']
```

```
print(df[df["job_id"] == 2169425610]["description"].to_list())  
# ['17th January 2021']
```

```
print(df[df["job_id"] == 2166953970]["description"].to_list())  
# ['Salt Lake City Metro Area, UT']
```

```
print(df[df["job_id"] == 2813519909]["description"].to_list())  
# ['Principals only. No agencies please.']
```

```
print(df[df["job_id"] == 2148292480]["description"].to_list())  
# ['Equal Opportunity Employer: All qualified applicants will receive consideration for  
employment without regard to race, color, religion, sex, sexual orientation, gender identity,  
national origin, disability, or status as a protected veteran.']
```

- No meaningful information related to the job at all.

Bad job postings: Only a link or an email

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```
print(df[df["job_id"] == 2151163725]["description"].to_list())  
# ['www.biontech.de']  
  
print(df[df["job_id"] == 2149693498]["description"].to_list())  
# ['https://j.brt.mv/ATS/PortalViewRequirement.do?reqGK=27415250']  
  
print(df[df["job_id"] == 2150975193]["description"].to_list())  
# ['https://www.linkedin.com/company/abalone-bio-inc']  
  
print(df[df["job_id"] == 2182318787]["description"].to_list())  
# ['careers@redpinrx.com']  
  
print(df[df["job_id"] == 2465425031]["description"].to_list())  
# ['To apply, please email your CV and cover letter to careers@biofidelity.com. ']
```

- Contain only a link or an email.

Bad job postings: Not enough information

► Back

```
print(df[df["job_id"] == 3605667513]["description"].to_list())  
# ['PhD - 1-4 Years of experience']  
  
print(df[df["job_id"] == 2299511922]["description"].to_list())  
# ["Bachelor's degree and 7 years of Scientific experience"]  
  
print(df[df["job_id"] == 3294469443]["description"].to_list())  
# ['Minimum 5 years of experience or equivalent education/experience']  
  
print(df[df["job_id"] == 2513185360]["description"].to_list())  
# ['Hplc, Gc']  
  
print(df[df["job_id"] == 2481383658]["description"].to_list())  
# ['Flow cytometry']  
  
print(df[df["job_id"] == 3306839629]["description"].to_list())  
# ['Familiarity with NGS workflows']
```

- Contain only part of the information (e.g., only education or skill requirements) we expect.

Evergreen job postings without an expected new hire

► Back

```
print(df[df["job_id"] == 2988280654]["description"].to_list())  
#:[ 'Sourcing Statement\n\nYou are applying to be part of the Novartis Talent Pool. We are not currently recruiting for this role but we are building a pipeline for future opportunities. If you would like to be considered for a similar position in future, then please submit your CV.\n\nJob Description\n\nThe Next Generation Scientist Program is an intensive training program for talented and motivated research scientists from low- and middle-income countries. The 8-week program is delivered virtually and consists of virtual classroom sessions, group work assignments and active individual contribution. The training curriculum is designed to foster both scientific and professional development of the post-graduate students. Guided by Novartis mentors, the participants learn about the pharmaceutical drug discovery and development process as well as participate in a leadership development program designed to enhance decision-making and communication skills. The program is run jointly by Novartis and the University of Basel.\n\nScientists with background and interest in malaria, sickle cell disease, leprosy or Chagas disease are encouraged to apply.\n\nApplication:\n\nPlease make sure to add the following documents to your application, combined in one PDF file:\n- CV or résumé\n- Letter of recommendation/ support from a faculty member\n- Letter of motivation how the program will benefit you\n\nCommitment to Diversity & Inclusion:\n\nNovartis is committed to building an outstanding, inclusive work environment and diverse teams representative of the patients and communities we serve.\n\nMinimum Requirements\n\nMinimum requirements\n- Enrolled at university as PhD or MD student in a geography of scope (low- and middle-income country)\n- Research background in malaria, sickle cell disease, leprosy or Chagas disease preferred\n- Active participation in all program activities expected\n- Good verbal and written English communication skills\n- Ability to work independently and in groups\n- Good organizational skills\n- Solid internet infrastructure for online training\n- Language: English\n\nDuration and dates: 8-week program from July to August 2022; three weekly classroom sessions (90min each)\n\nPreferred field of studies:\n\nNatural and Pharmaceutical Sciences, Medicine, and related fields of study\n\nWhy Novartis?\n\n799 million. That's how many lives our products touched in 2019. And while we're proud of that, in this world of digital and technological transformation, we must also ask ourselves this: how can we continue to improve and extend even more people's lives?\n\nWe believe the answers are found when curious, courageous and collaborative people like you are empowered to ask new questions, make bolder decisions and take smarter risks.\n\nWe are Novartis. Join us and help reimagine medicine.\n\nDivision\n\nPHARMA\n\nBusiness Unit\n\nGlobal Health\n\nPH\n\nCountry\n\nSwitzerland\n\nWork Location\n\nBasel\n\nCompany/Legal Entity\n\nNovartis Pharma Basel\n\nFunctional Area\n\nResearch & Development\n\nJob Type\n\nFull Time\n\nEmployment Type\n\nInternship\n\nShift Work\n\nNo\n\nEarly Talent\n\nYes']
```

- Explicitly mention that “We are not currently recruiting for this role ...”

The sample of candidate job postings

► Back

- 1 Construct one extraction window per focal company. For each exact company identifier, extend from 18 months before its first focal hire through 3 months after its last focal hire.
- 2 Retain job postings within those companies that have at least 1 focal new hire.
- 3 Retain job postings within the same focal ONET occupations.
- 4 Retain job postings within the Step-1 constructed company-specific posting-date windows.
- 5 Exclude internship advertisements by specified job titles.

The sample of candidate informative job postings

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- 1 Hash exact descriptions so repeated text is processed once, while retaining every posting in the final sample.
- 2 Parse HTML or plain text into ordered blocks, normalize the text conservatively, and hash each distinct normalized description.
- 3 Keep a description only if it has at least 600 visible characters, 100 words, three informative units, two substantive content areas, and duties or qualification content. Exclude boilerplate-dominated, corrupted, noncurrent, multi-role, or truncated text.
- 4 Map the description-level decision back to every posting. This identifies 240,464 informative postings, or 60.8% of the 395,488 US candidate postings.

A potential weighting scheme based on Poisson assumptions

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For new hire i and candidate posting $j \in \mathcal{C}_i$, let d_{ij} be the number of months from the posting date to the hire's start date (our proxy for the vacancy's fill date).

- Suppose acceptable matches arrive according to a homogeneous Poisson process with rate λ . The time to the first match is exponential (assume on average it takes 2 months to fill in a vacancy):

$$T \sim \text{Exp}(\lambda), \quad \mathbb{E}[T] = \frac{1}{\lambda} = 2 \quad \implies \quad \lambda = \frac{1}{2} \text{ per month.}$$

- The temporal likelihood that posting j generated hire i is therefore $L_{ij} = \lambda \exp(-\lambda d_{ij})$ for $d_{ij} \geq 0$.
- Given a prior match probability π_{ij} (e.g., based on title or geography), normalize within the candidate set:

$$w_{ij} = \frac{\pi_{ij} L_{ij}}{\sum_{k \in \mathcal{C}_i} \pi_{ik} L_{ik}} = \frac{\pi_{ij} \exp(-d_{ij}/2)}{\sum_{k \in \mathcal{C}_i} \pi_{ik} \exp(-d_{ik}/2)}, \quad \sum_{j \in \mathcal{C}_i} w_{ij} = 1.$$

- With equal priors, only timing matters. Each additional month multiplies a posting's relative likelihood by $\exp(-1/2) \approx 0.607$. For lags of 1, 3, and 6 months, the normalized weights are approximately 0.690, 0.254, and 0.057.

Outline

④ Appendix: Supplementary materials

⑤ Appendix: References

References I

- Deming, David and Lisa B. Kahn (2018). “Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals”. *Journal of Labor Economics* 36.S1, S337–S369.
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- Gnehm, Ann-Sophie and Simon Clematide (2020). “Text Zoning and Classification for Job Advertisements in German, French and English”. *Proceedings of the Fourth Workshop on Natural Language Processing and Computational Social Science*. Online: Association for Computational Linguistics, pp. 83–93.
- Green, Thomas, Diana Maynard, and Chenghua Lin (2022). “Development of a Benchmark Corpus to Support Entity Recognition in Job Descriptions”. *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. European Language Resources Association, pp. 1201–1208.
- Krüger, Kai, Johanna Binnewitt, Kathrin Ehmann, Stefan Winnige, and Alan Akbik (2025). “Improving Online Job Advertisement Analysis via Compositional Entity Extraction”. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Suzhou, China: Association for Computational Linguistics, pp. 27035–27053.
- Nguyen, Khanh, Mike Zhang, Syrielle Montariol, and Antoine Bosselut (2024). “Rethinking Skill Extraction in the Job Market Domain Using Large Language Models”. *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*. St. Julian’s, Malta: Association for Computational Linguistics, pp. 27–42.

References II

Zhang, Mike, Kristian Jensen, Sif Sonniks, and Barbara Plank (2022). “SkillSpan: Hard and Soft Skill Extraction from English Job Postings”. *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Seattle, United States: Association for Computational Linguistics, pp. 4962–4984.